Research Proposal & Execution Plan

Consistency-Constrained Multi-Task Bengali Hate Speech Detection with Faithfulness-Evaluated Generative Explanations

Research Strategy Document

August 2026

Abstract

This document outlines the overarching goals, real-world applications, methodology, and step-by-step execution guide for our original research project targeting the ICCIT 2025 conference. It serves as a master blueprint for the project.

1 1. The Discovered Gaps & Our Proposed Goal

Through rigorous literature review (including verifying the distinction between the BanHate classification dataset and the BanHADEX explainability dataset), we have verified three critical gaps in Bengali hate speech detection:

G1. No Model-Level Consistency Constraint: Current Multi-Task models (like Code_Gen or HateSense) operate independent heads, leading to logically contradictory outputs (e.g., predicting "No Hate" alongside "Severe Toxicity").

G2. No Generative Faithfulness Evaluation: While models can generate fluent Bengali rationales, nobody has evaluated if these explanations actually reflect the model's true reasoning using ERASER-style metrics (Comprehensiveness/Sufficiency).

G3. No Unified System: No system successfully combines Multi-Task dimensional predictions with faithful generative explainability.

Our Proposed Goal: We propose a **Consistency-Constrained Multi-Task Bengali Hate Speech Detection framework with a Faithfulness-Evaluated Generative Explanation pilot**. To ensure feasibility for ICCIT 2025, our primary contribution will strictly focus on mathematically solving **G1** (the Consistency Constraint). As a secondary, highly novel pilot contribution, we will address **G2/G3** by evaluating the faithfulness of generated explanations on a curated subset of the data.

2 2. Real-World Applications

If successful, this research has profound implications and highly practical applications:

- **Trustworthy Content Moderation (Meta/YouTube/TikTok):** Social media companies operating in Bangladesh desperately need automated tools. A system that can not only detect hate but also provide a *guaranteed faithful* explanation in Bengali will massively improve user trust in automated takedowns.
- **Reducing Human Trauma:** By confidently and consistently flagging severity and targets with explanations, human content moderators will spend less time reviewing highly toxic content, protecting their mental health.

- **Transparent Policy Enforcement:** Governments and regulatory bodies requiring AI transparency (e.g., AI safety acts) will mandate models that do not output contradictory labels and can logically defend their classifications.

3 3. Proposed Methodology (What We Will Do)

To achieve our goals, we will develop a novel neural network architecture based on PyTorch.

Core Architecture Components

- **Shared Encoder:** We will use a pre-trained transformer (e.g., `BanglaBERT`) to extract rich contextual embeddings from Bengali social media text.
- **Multi-Task Heads:** We will attach three distinct classification heads to predict Hate Type (6 classes), Target (5 classes), and Severity (3 classes).
- **Consistency Penalty Loss (Novelty 1):** We will design a custom loss function that heavily penalizes the model during training if it predicts logically impossible label combinations across the three heads.
- **Generative Rationale Head (Novelty 2):** We will attach a Sequence-to-Sequence (Seq2Seq) decoder that generates a free-text Bengali explanation justifying the classification.
- **Faithfulness Evaluation:** We will evaluate the generative head using ERASER metrics (Comprehensiveness and Sufficiency) to prove the model isn't hallucinating.

4 4. Feasibility & Detailed 20-Day Execution Plan

Feasibility Assessment: Compute is not our bottleneck (Kaggle's free tier provides ∼30 GPU hours/week, which is comfortably enough for fine-tuning BanglaBERT). *Time* is the constraint. We cannot realistically build three novelties, reproduce three SOTA baselines from scratch, and merge datasets perfectly in just 20 days.

Therefore, we have enacted a **Strategic Scope Cut**:

- We will NOT train Code_Gen or HateSense from scratch; we will simply cite their published numbers as baselines.
- The generative explainability (Faithfulness) will be framed as a "preliminary pilot study" on a subset of data (200-500 samples), while the Consistency Constraint will be the fully robust, primary headline contribution.

Here is our scoped-down, highly realistic 20-day execution plan:

Days 1-2: Setup & The "Merge" Problem

- Confirm BanglaMultiHate dataset access on Kaggle.
- Set up a local API script (using Gemini/OpenAI) to generate silver-standard Bengali rationales for a *small pilot subset* (500 samples) of the training data.

Days 3-10: Primary Contribution (Consistency Constraint)

- **Days 3-5:** Train the baseline single-task and vanilla multi-task BanglaBERT models. Log per-task F1.
- **Days 6-8:** Implement the novel **Consistency Penalty Loss**. Run sweeps and compute the *Contradiction Rate*. This forms the headline result table of the paper.
- **Days 9-10:** Perform ablations (turning Focal loss on/off, consistency on/off).

Days 11-15: Secondary Contribution (Faithfulness Pilot)

- Train the lightweight Seq2Seq rationale generator on the 500-sample pilot subset.
- Execute ERASER metrics (Comprehensiveness and Sufficiency).
- Extract highly compelling qualitative examples showing where the model succeeds and hallucinates.

Days 16-20: Writing the Manuscript

- Fill in the remaining sections of our LaTeX manuscript (`iccit_paper_2025.tex`) with actual numerical results.
- Trim the Abstract and Introduction to match exactly what was built (Primary Consistency + Pilot Explainability).
- Format to the strict IEEE 6-page limit and submit.